Fiscal Year:	FY 2020	**Task Last Updated:**	FY 11/24/2020
PI Name:	Papier, Arthur M.D.		
Project Title:	VisualDx Clinical Decision Support for Ultrasound		
Division Name:	Human Research		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	TRISH--TRISH		
Joint Agency Name:		**TechPort:**	Yes
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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City:	Rochester	**State:**	NY
Zip Code:	14604	**Congressional District:**	25
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	TRISH--Industry
Start Date:	05/01/2019	**End Date:**	07/31/2020
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	TRISH
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:	NOTE: End date changed to 7/31/2020 (original end date was 4/30/2020) per TRISH (Ed., 6/18/2020)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:	NNX16AO69A-IND0009		
Performance Goal No.:			
Performance Goal Text:			
Task Description:	Industry Project Spaceflight on-board medical diagnostic systems need to be intuitive to minimize training time and point-of-care ultrasound is a good fit for this need. VisualDx proposed improving their ultrasound system by incorporating clinical decision support to help non-specialists make accurate diagnoses. First, a set of annotated images will be curated, showing disease conditions, including appropriate variations. Next, space-specific guidance will be expanded, and a work-up module will be developed that incorporates image-based pattern matching and questions regarding symptoms. These features will allow non-specialists with limited ultrasound training to identify potential diagnoses and deliver the best care.		

Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	As part of a previous project, VisualDx was able to provide a medical decision support tool that can function on deep-spaceflight missions without access to internet. The previous project included over 100 conditions on the Exploration Medical Condition List (MCL) and the product was built to support the addition of more conditions and modules moving forward. The current project is one such expansion into ultrasound content. Point of care ultrasound is a well-accepted and well-developed mode of diagnostic imaging that requires minimal training for image acquisition, but does require more in-depth knowledge or support to properly interpret the results. By adding an abdominal ultrasound module to the existing VisualDx product, we can now support non-expert astronauts in their medical decision-making related to diagnosing themselves or others using ultrasound devices during spaceflight. On Earth, the new ultrasound-specific module and content allows VisualDx to continue to expand its applicability in the global health context in the developing world and other under-resourced environments. As the use of portable ultrasound devices grows in these environments, the standalone version of VisualDx is now well-positioned to support lesser-trained health care providers in interpreting the ultrasound results to make better decisions and provide better care. Further, as part of this project, VisualDx created a hierarchy of ultrasound findings for high-yield space-specific abdominal conditions, which in theory can be expanded to other conditions throughout the body that can be diagnosed with ultrasound. To our knowledge, this is the first formally defined hierarchy of ultrasound findings, and the knowledge base being built into this structure can be used to support further VisaulDx content expansion that could eventually have even more impact both in Space and on Earth.
Task Progress:	VisualDx is the most widely used diagnostic support system in the world. VisualDx combines machine learning and vision science with structured clinical data and clinical question sets to let non-specialist healthcare providers (1) enter patient specific findings, (2) build custom differentials, and then (3) view images, treatment recommendations, and management information. Spaceflight on-board medical diagnostic systems need to be intuitive to minimize training time and point-of-care ultrasound is a good fit for this need. With previous TRISH (Translational Research Institute for Space Health) project funding, VisualDx had developed a standalone (no internet required) version of its clinical decision support tool and for this project, proposed to expand that tool by incorporating a new workflow and content to help non-specialists make accurate diagnoses based on point-of-care ultrasound findings. This project allowed VisualDx to enhance its existing diagnostic capabilities by creating an ultrasound-specific case workup module, as well as expanding representation of ultrasound imaging to include multiple annotated images for a targeted list of conditions likely to occur in space. These features will allow non-specialists with limited ultrasound training to identify potential diagnoses and deliver the best care.
Bibliography Type:	Description: (Last Updated:)